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2024 Oka Stream CRD Summary Report

Rev1.0, CQ&R CMEE Team, 2024 Q4



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Objectives

Share Findings From CMEE Server Industry Data Collection (Oka Stream Platform Customer Survey)

- Data Collected From 7 OEM/ODM Customers/Sites
 - Bytedance
 - Foxconn Tianjin
 - IEIT (Inspur)
 - Inventec TW
 - Lenovo/LCFC
 - Quanta TW
 - xFusion

Highlights

- Customer common stencil thickness is 5~6mil ➔
- Alpha, Eunow and Shenmao are commonly used paste ➔
- Ersa and Heller are mainstream reflow oven ➔
- Highest peak reflow temperature can up to 250~255C ➔
- 4 customers have N2 rework capability ➔
- 450ue still the most commonly spec customer can meet ➔
- Most customers direct copy Intel reference board design land pattern ➔

Solder Paste Print

Solder Paste Printing (1 Of 4)

	Paste Print Machine	Squeegee Speed	Squeegee Pressure	Stencil Separation Speed
Bytedance/H3C	GKG K870	65 mm/sec	10 +/-3kg	3 ~10mm/s
Foxconn	GKG Pmax-6 DEK 03IX	50 +/-10mm/sec	12 +/-2kg	0.5mm/s
IEIT L6 (Inspur)	GKG Pmax-8	80 +/-30mm/sec	13 +/-2kg	0 ~3mm/s
Inventec	Fuji GPX	20~60 mm/sec	80 ~ 160N (8.2~16.3kg)	0.05 ~ 0.6mm/s
Lenovo/LCFC	No Answer	No Answer	No Answer	No Answer
Quanta	Fuji GPX-CII	45mm/sec	150N (15.3kg)	0.2mm/s
xFusion	No Answer	No Answer	No Answer	No Answer

Solder Paste Printing (2 Of 4)

	Under Screen Cleaning Frequency	Stencil Cleaning Frequency	Cleaning Process	PCB Support Type
Bytedance/H3C	Every 1 PCB	Every 2 hours	Ultrasonic	Vacuum & block
Foxconn	Every 2 PCB	Every 12 hours	Spraying	Support block
IEIT L6 (Inspur)	Every 1 PCB	Every 8 hours	Cleaning paper: Wet wipe & Dry clean & Dry wipe	Support pin, Support block
Inventec	Every 1 ~ 3PCB	No answer	No answer	Dedicate support block
Lenovo/LCFC	No Answer	No Answer	No Answer	No Answer
Quanta	Every 1 PCB	Every 12 hours	Ultrasonic	Dedicate support block
xFusion	Every 1 PCB	Every work order	Ultrasonic	Support pallet

Most customer do under stencil clean after 1~2 prints

Solder Paste Printing (3 Of 4)



	Do you use the same pallet for paste print, PnP and reflow?	Maximum board handling capability (length x width)	Stencil Technology + coating	Stencil Thickness Used
Bytedance/H3C	Yes	870 x 610 mm	Laser cut with electro polish + nano coat	5 mils
Foxconn	No, pallet only for reflow	850 x 610 mm	FG + laser cut with electro polish	4 mils
IEIT L6 (Inspur)	Yes	850 x 610 mm	Electroform, laser cut with electro polish + nano coat	4 ~ 6 mils
Inventec	No	21" x 24" (533.4 x 609.6 mm)	FG, laser cut with electro polish + nano coat	5.91 mils
Lenovo/LCFC	No answer	No answer	FG, laser cut + nano coat	5 mils
Quanta	Yes	610 x 610 mm	laser cut with electro polish + nano coat	6 mils (skt), 4 mils (BGA)
xFusion	Yes	800 x 800(?) mm	laser cut with electro polish + nano coat	0.12 mm (~4.7 mils)

Common stencil thickness 5~6 mils

Solder Paste Printing (4 Of 4)



	Is a step stencil used?	PCB surface finish & thickness	What type of PCB is used?	Solder Paste used
Bytedance/H3C	Yes, base is 0.12mm, with step down to 0.1mm, or step up to 0.15mm, 0.18mm	OSP, 118 mils	Type 4	Murata, Eunow, Tongfang Type 4
Foxconn	No answer	OSP, 62 mils, 98 mils, 118 mils	Type 12 (?)	Alpha OM338PT, ShenMao PF606-P, Type 4
IEIT L6 (Inspur)	Yes, base 4mils, step up 1-2mils	ENIG & OSP, 1.0~3.8 mm (39 mils ~ 150 mils)	No answer	ShenMao, Alpha, Eunow Type 4
Inventec	Yes	OSP, 100 mils	No answer	Type 4
Lenovo/LCFC	Generally, it is about at least 0.5mm greater than or equal to the area where the part is located.	OSP, 62 ~ 118 mils	No answer	Type 4
Quanta	No answer	Most OSP, some ENIG	Type 4	ShenMao PF606-P26, Type 4
xFusion	Socket will be step up	No answer	No answer	Type 4

Customer capable of using step stencil; ShenMao, Eunow and Alpha are commonly used pastes

Pick and Placement (PnP)

Pick and placement (1 Of 1)

	PnP Machine	Nozzle size (LGA socket & Intel component)	Max component size capability (x, y)	Max component weight capability (grams)	Min pitch/ball recognition capability
Bytedance/H3C	Fuji NXT M6	20G	200 x 150 mm	260 g	0.3 mm pitch
Foxconn	Panasonic NPM-W2 USI FUZION 2-14	20x20mm(NPM 3451) 16.5x16.5mm(FUZION H065)	150 x 150mm (FUZION) connector: 100(L) x 90mm(W) (NPM)	300g (NPM) 200g (FUZION)	0.094/0.118mm(0.94mil camera)(FUZION) 0.234/0.292mm(2.3mil camera) (FUZION) 0.5/0.5(BGA) 0.25(CSP)mm (NPM)
IEIT L6 (Inspur)	Panasonic NPM-W2	1005	L 120 x W 90 mm or L 150 x W 25 mm	168 g	0.4 mm
Inventec	No answer	10~20 mm	74 x 74 mm(140 x 32) normal 150 x 150mm (option)	160 g / 80 g option / normal	0.4 mm pitch / 10000 balls
Lenovo/LCFC	No answer	24 mm(outside) / 21.5 mm (inside)	200 x 110 mm	160 g	0.1 mm
Quanta	Fuji AIMAX3	20 mm	135 x 135 mm	250 g	Min 0.8 mm in mess production. Machine capable smaller
xFusion	No answer	No answer	No answer	No answer	No answer

Main PnP machines are Fuji and Panasonic

Reflow

Reflow (1 Of 3)



	Reflow Oven	Reflow support	Use N2?	O2 PPM
Bytedance/H3C	Ersa HOTFLOW 3/26XL	Yes	Yes	< 1000 PPM
Foxconn	Ersa HOTFLOW 3/26XL	Yes	Yes	< 1000 PPM
IEIT L6 (Inspur)	Heller 1809exl Ersa HOTFLOW 3/26XL	Yes	Yes	< 500 PPM
Inventec	Heller 1809exl	Yes	Yes	< 1000 PPM
Lenovo/LCFC	Heller 1809exl	Yes (picture)	Yes	No answer
Quanta	Kurtz ersa	No answer	Yes	< 2000 PPM
xFusion	Heller 1809exl	Yes, aluminum pallet	Yes	< 3000 PPM

Ersa and Heller are mainstream reflow oven

Reflow (2 Of 3)



	Belt Speed	Ramp Rate	Cooling Rate	Peak Temp Range	Time above Liquidus
Bytedance/H3C	95 cm/min	2.5 °C/s	4 °C/s	230-250 °C	45-90 seconds
Foxconn	20-200 cm/min	0-2.5 °C/s	1.5 °C/s	235~255 °C	60-120 seconds
IEIT L6 (Inspur)	30-180 cm/min	1-3 °C/s	1-3 °C/s	230-250 °C	60-90 seconds
Inventec	78 cm/min	< 3 °C/s	< 2 °C/s	237-255 °C	65-115 seconds
Lenovo/LCFC	80 cm/min	< 3 °C/s	< 2 °C/s	230-250 °C	60-90 seconds
Quanta	Depends	1~3 °C/s	1~3 °C/s	235-250 °C	60-90 seconds
xFusion	No answer	No answer	No answer	No answer	No answer

Highest peak reflow temperature can be up to 250~255C

Reflow (3 Of 3)

	Soak Temp Range and Time	What is the delta T across the component	What is the number of heating and cooling zones used?	Maximum board handling capability (length x width)	SMT Yield Target
Bytedance/H3C	No answer	5 °C	9 ramp, 4 heating/reflow, 4 cooling zones	1068mm x 610mm	1000 DPM
Foxconn	Soak Time (145-175°C): 60-120 seconds	< 5 °C	13 heating/reflow, 4 cooling zones	680mm x 610mm	No answer
IEIT L6 (Inspur)	60-120 seconds	< 10 °C	9 ramp, 4 heating/reflow, 4 cooling zones	850mm x 610 mm	99.5% (whole process FPY)
Inventec	55-125 seconds	< 10 °C	4 ramp, 5 heating, 4 reflow, 4 cooling zones	534mm x 610mm	No answer
Lenovo/LCFC	90-120 seconds	< 10 °C	4 heating & cooling zones	500mm x 600mm	No answer
Quanta	90-120 seconds	< 10 °C	9 ramp, 4 heating/reflow, 4 cooling zones	540mm x 600mm	98.5% (whole process FPY)
xFusion	No answer	No answer	No answer	800mm	Depends

600+ mm is the max board width customer can handle

Rework and Others

Rework and Others (1 Of 2)

	Rework Machine	Peak Temp Range	Time above Liquidus	Soak Temp Range and Time	What size of spot heater do you have/use?
Bytedance/H3C	VJ Summit Lxi	230-250 °C	No answer	No answer	125mm x 125 mm
Foxconn	No answer	235-250 °C	60-120 seconds	60-120 seconds	120mm x 120 mm
IEIT L6 (Inspur)	No answer	230-250 °C	60-90 seconds	60-120 seconds	76mm x 76mm
Inventec	No answer	235-250 °C	60-150 seconds	60-150 seconds	480mm x 480mm
Lenovo/LCFC	No answer	235-250°C	60-120 seconds	150-200°C/60-120S	According to part specifications
Quanta	No answer	235-250 °C	60-90 seconds	90-120 seconds	70mm x 70mm
xFusion	No answer	No answer	No answer	No answer	50mm x 50mm

Rework and Others (2 Of 2)



	Rework Paste/Flux	Rework use N2?	Rework Yield Target	Do you have MSL-6 handling BKM's?	What are your FA capabilities? (DnP, XS, etc.). XS size restrictions?
Bytedance/H3C	Murata, Eunow	Yes, O2 < 1000 PPM	99%	Yes	Yes, XS size < 150mm
Foxconn	No answer	No	No answer	No answer	No answer
IEIT L6 (Inspur)	No answer	Yes	99.9%	No	Yes
Inventec	No answer	Both	No answer	Yes	DnP, XS, shadow moiré, push-pull
Lenovo/LCFC	No answer	Yes	95%	Yes	Yes
Quanta	No answer	No	99.9%	No answer	No answer
xFusion	No answer	No	No answer	No answer	No answer

4 customers have N2 rework capability, 3 customers didn't provide answer

Automation

Automation (1 Of 1)

	Do you use automation in your assembly process?	Do you have Full or partial automation during the server assembly process. If partial, which part of the assembly process is automated?	Does the automated test or assembly occur in-line or off-line?	If automation is used, what type of equipment/machines are used?	What is used to pick & place Intel's product or retention components (piece parts) ?
Bytedance/H3C	Yes	Except cable assembly, all others are automated	In line	Robotic arm	vacuum nozzle for socket, gripper for piece parts
Foxconn	No answer	Only auto screw for bolster plate	In line	Auto screw machine	Manual
IEIT L6 (Inspur)	Yes	Auto screw, PCBA pick and place	In line	Self design automation	Gripper
Inventec	Not able to share	Not able to share	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	No	No	In line	No	No
Quanta	Yes	Test – full, Assembly - partial	In line	No answer	Gripper and nozzle
xFusion	Yes	Test – full, Assembly - partial	In line	Auto inspection, auto screw	No answer

Assembly Process

Assembly (1 Of 1)

	Do you use a fixture for the assembly of the PHM components? If yes, what kind of fixture	Do you follow the MAS for the PHM assembly and torque sequence?	What torque is used for the HS assembly?	What RPM is used for heatsink nut torque driver?	Do you have a lighting requirement for the assembly line?
Bytedance/H3C	Yes, use PHM assembly fixture	Yes	8 in-lbf	No specification	Yes, follow GB5700
Foxconn	Yes	Yes	8 in-lbf	800 rpm	860~1200 lux
IEIT L6 (Inspur)	Yes	Yes	8 ~ 10 kgf-cm (6.9 ~ 8.7 in-lbf)	1000 rpm	750 lux for assembly 1200 lux for inspection
Inventec	Not able to share	Not able to share	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	Yes	Yes	8 in-lbf	1000 rpm	800 lux
Quanta	Yes	Yes	For LGA4677, we use 8.5 kgf-cm (7.4 in-lbf)	No specification	600 lux
xFusion	No answer	No answer	3~6 kgf (6.6~13.2 lbf)	No answer	No answer

8 in-lbf torque is most common used for HS assembly

Others for System Integration

Others for System Integration (1 Of 2)

	What area of improvement would you like to see?	What heatsink design is used?	Do you have experience with RMHS (remote module heatsink) or ganged heatsinks (2 attached HS)?
Bytedance/H3C	None	Intel reference design	Yes, installed by mechanical arm to ensure flatness
Foxconn	None	Own designed heatsink, liquid cooling	No
IEIT L6 (Inspur)	Hope to improve socket dust cover to be easy install/remove by hand, and reduce bend contact	Use own designed heatsink with optimization, reduced assembly complexity and change to cause bent contact	Have experience on ganged heatsinks (2 attached HS)
Inventec	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	Hope to improve the flatness of CPU bracket	Intel reference design	No
Quanta	None	EVAC	Yes , the PHM module is designed by Quanta ME team , it will install on CPU and connect with others via cooper pipes, but module is all in one type .
xFusion	Hope to improve socket dust cover to be easy install/remove by hand	Air cooling and cold plate cooling	Yes, RMHS (remote module heatsink)

**2 customers hope to improve dust cover design, 1 customer want to improve CPU bracket flatness;
4/7 customers have experience on RMHS or ganged HS**

Others for System Integration (2 Of 2)

	What TIM material is used?	Do you use the Intel designed socket cap removal tool?	Are different products assembled on the same assembly line?
Bytedance/H3C	Thermal grease	Yes	Yes
Foxconn	Thermal grease	No, we use our own designed tool	Yes
IEIT L6 (Inspur)	PTIM, thermal grease, thermal pad	Yes	Yes
Inventec	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	Thermal pad	Yes	Yes
Quanta	Thermal pad	No, we use our own designed tool	For mass production, the product line will be fixed
xFusion	Thermal grease	No answer	Yes

Manufacturing Test Process

Manufacturing Test Process (1 Of 1)

	Are ICT testers used? If yes, please provide ICT tester model	How many times is assembly test done before thermal solution attach?	How are boards transported between all assembly and test process steps?
Bytedance/H3C	Yes, Teradyne test station LX/LH	4 (AOI,AXI,ICT,FBT)	Lay flat in a tray, picture
Foxconn	Yes, Keysight 3070	3 (MDA, ICT, BSI), for reworked board, total 6 times	Line conveyor and AGV (Automated Guided Vehicle)
IEIT L6 (Inspur)	Yes, Teradyne TSLX/TSLX2 or flying probe SPEA 4060S2	1 (ICT)	Lay flat in a tray, move by cart. picture
Inventec	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	Yes, Agilent 3070, Teradyne, and TRI 5001	1 (ICT)	Using a cart, place the motherboard vertically on the cart (or horizontally in a tray)
Quanta	Yes, Keysight	1	When the PCB need to be repaired, we use carts to transfer, other we use conveyor. picture
xFusion	Yes	2 (repair board 4 times)	Cart or tray, lay flat

Board Strain Monitoring

Board Strain Monitoring (1 Of 3)



	When do you monitor board strain through board assembly, test, and handling steps?	What board strain guidance do you use?	What is the lowest BFI strain that you have been able to meet through board assembly, test, handling steps prior to thermal-solution attach?	How many total process steps (assembly, test, handling) do you monitor board strain on?
Bytedance/H3C	For every project	Follow Intel BFI guidance	< 450ue	1. ICT test; 2. De-panel; 3. Functional test; 4. Thermal module assembly; 5. System assembly;
Foxconn	For every project	Follow customer request	< 450ue	Same as above
IEIT L6 (Inspur)	For every project	No answer	< 450ue	1. ICT test; 2. De-panel; 3. Functional test; 4. Thermal module assembly
Inventec	Not able to share	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	For every project, every fixture	Follow Lenovo spec	< 450ue	1. ICT test; 2. De-panel; 3. Functional test; 4. Thermal module assembly
Quanta	For every project	Follow CPU vendor guidance	< 250ue	1. ICT test; 3. Functional test; 4. Thermal module assembly; 5. System assembly; - no de-panel
xFusion	No answer	No answer	No answer	No answer

450ue still the most commonly spec customer can meet

Board Strain Monitoring (2 Of 3)

	Do you collect any solder joint FA (Failure Analysis) after board assembly, test, and handling steps?		
	If yes, what type of tools are used?	If yes, when?	If yes, what is the purpose?
Bytedance/H3C	DnP, XS, X-Ray	For every project	as a pre-step for board or system-level reliability testing
Foxconn	DnP, XS, X-Ray	Follow customer's test matrix	1. as a pre-step for board- or system-level reliability testing (e.g. TC or shock) 2. as a board or system assembly time=0 solder joint quality control check
IEIT L6 (Inspur)	DnP, XS, X-Ray	For every project	Follow customer's request
Inventec	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	DnP, XS	1. Introduce new project 2. When introduce new material, for example: Skt, DIMM, new PCB material	1. as a pre-step for board- or system-level reliability testing (e.g. TC or shock) 2. as a board or system assembly time=0 solder joint quality control check
Quanta	DnP, XS	For every project and every stage	Same as above
xFusion	No answer	No answer	No answer

Board Strain Monitoring (3 Of 3)

	What is the largest BGA package and socket that you can currently manufacture through board assembly, test, and handling steps?	Do you anticipate any challenges controlling board strain as packages grow larger, more rectangular, etc.?	Would you be willing to share BFI strain data that you have collected on an Intel product?
Bytedance/H3C	125 x 125 mm	Yes, test fixture design and handling will have lower margin	No
Foxconn	150 x 150 mm	Yes, test fixture design and handling will have lower margin	Yes. We can share one
IEIT L6 (Inspur)	135 x 135 mm	Yes, 1. solder joint quality (open/short), 2. Lower flatness, 3 hard to rework	Yes
Inventec	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	200 x 125 mm	Yes, the CPU bracket flatness is hard to control, which will impact strain gauge test	No
Quanta	70 x 73 mm (in HVM)	Yes	No, it's customer's confidential file
xFusion	No answer	No answer	No answer

Socket Land Pattern

Socket Land Pattern(1 Of 1)



	Do you use Intel's reference board design archive or the land pattern guidance document to create your CPU socket library symbol?	Do you build your socket symbol with soldermask compensation (soldermask opening larger than metal defined pads), non-compensated (soldermask the same size as metal defined pads), or a combination of both?	How does the usability of the collateral provided by Intel for the physical symbol for server sockets compare to other suppliers of similar sockets?	How are motherboard mounting hole locations determined?
Bytedance/H3C	No answer	A combination of both	No answer	Depend on structure model/drawing to decide mounting hole location
Foxconn	No answer	No answer	No answer	No answer
IEIT L6 (Inspur)	Used & copied Intel's reference board design with library symbol for socket land pattern	Non-compensated (soldermask the same size as metal defined pads)	Based on the current experience of PR and MDP projects, Intel directly provides library/symbol usage and design guidance documents, which is better for design and development.	Used & copied Intel's reference board design
Inventec	Not able to share	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	Direct copy from Intel reference brd to create own symbol file	Non-compensated (soldermask the same size as metal defined pads)	Intel's symbol file is better than competitors'	Direct copy from Intel reference brd
Quanta	Follow the Intel recommendation	A combination of both	No answer	No answer
xFusion	Copy reference board design for socket land pattern	Copy reference board and improve based own experience	Intel provides more detail information	Skt mounting hole copy Intel, others depend on board stress simulation

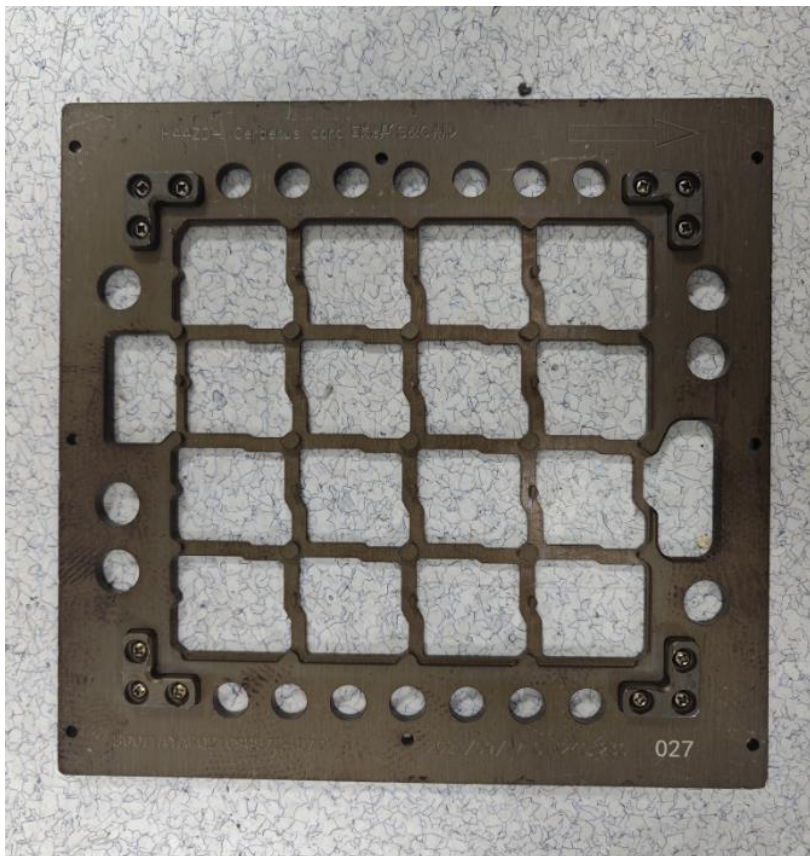
Board Reference Design

Board Reference Design(1 Of 1)

	Do you follow Intel's reference design closely under Intel components (i.e., under the socket)? 1. Via Type and size 2. Routing and power shapes 3. routing configuration 4. dielectric thickness around high speed signal layers	Do you utilize the PCB materials validated by Intel's reference design for boards using Intel components?	Is there additional information about manufacturing related aspects of the reference board in areas under Intel components that is needed?
Bytedance/H3C	Yes	We reserve 4 kinds PCB material, and select from them based on SI performance, cost, availability etc.	Hope MAS can be provided earlier
Foxconn	No answer	No answer	No answer
IEIT L6 (Inspur)	Used & copied Intel's reference board design	Utilize the PCB materials validated by Intel's reference design.	For board design, used & copied Intel's reference board design first, requires Intel to provide the official PDG.
Inventec	Not able to share	Not able to share	Not able to share
Lenovo/LCFC	Refer to CRB	Utilize the PCB materials validated by Intel's reference design	Can Intel provide VIPPO or SKIPVIA methodology and back drill depth?
Quanta	No answer	No answer	No answer
xFusion	Refer to Intel reference design, and perform in depth SI/PI optimization	Depends, we have our own certified PCB material	For big package, the process window is small, hope Intel can provide gen to gen changes of DFM rules, CTE requirements, board thickness, layer to layer alignment requirements etc.



Reflow Pallet



LCFC Reflow Pallet

GB5700



8.2照明

8.2.1参数标准

参照GB5700《室内照明监测方法》，结合实际状况。

1) 照度基本要求定义：

视觉作业特性	识别对象的最小尺寸d (mm)	照度范围 (lx)	
		一般照明	混合照明
特别精细作业	$d \leq 0.15$	—	1500-2000
很精细作业	$0.15 < d \leq 0.3$	200-300	750-1000
精细作业	$0.3 < d \leq 0.6$	150-200	500-750
一般精细作业	$0.6 < d \leq 1.0$	100-150	300-500
一般作业	$1.0 < d \leq 2.0$	75-100	200-300
较粗糙作业	$2.0 < d \leq 5.0$	50-75	—
粗糙作业	$5.0 < d$	30-50	—
一般观察生产过程	—	10-20	—
大件储存	—	5-15	—

注：

一般照明：是指不考虑特殊的局部需要，为整个照明区域提供大致相等的照明水平。

局部照明：是为了增加某些特定区域的照度而设置的。

混合照明：是指一般照明和局部照明相结合的整体照明方式。

ByteDance ODM Board Transportation



IEIT Board Transportation



Quanta Board Transportation

